there arises the question, why a nucleus in which all the prerequisites of a transmutation are given, does not change immediately.

The differences in the lifetimes of the various beta emitters are smaller than those among the alpha emitters. The half-lives of beta emitters vary from a few seconds to a few years. Only very few of them have a much longer half-life.

We cannot account for beta radiation by the same arguments which we used to explain the properties of alpha emitters; at any rate, these arguments are not applicable to the electron emitters, for every electron carries a negative charge, and consequently, when outside of the nucleus, it is attracted by the latter. Therefore, in this case, the potential barrier is non-existent. However, since there is no essential difference between electron emitters and positron emitters, we cannot suppose that there is an analogy between positron emitters and alpha emitters either. Furthermore, there is the fact that the electrons and pcsitrons, together with their accompanying neutrinos, do not constitute integral parts of the nucleus as do the alpha particles, but are created from the nuclear field only at the instant when the nuclear transmutation actually occurs. An analogy with the emission of photons from the extranuclear atomic structure seems to be a more logical one.

Empirically, it is generally true that the decay probability of the beta emitters also increases with the energy of the beta particle. It must be borne in mind that every electron and positron is accompanied by a neutrino, and as already pointed out, the reaction or decay energy is shared by this pair according to statistical laws. Therefore, the energy of the fastest beta particle, the neutrino of which happened not to receive any share of the decay energy, is decisive for the decay energy.

Now, in order to understand, at least qualitatively, the relationship between energy and decay probability, let us borrow from the theory of electric waves. For a beta radiation should actually be likened to the emission of light from the extranuclear atomic structure. We are thinking now in terms of the wave aspect, and therefore will speak of an electron or a positron, plus a neutrino, as a wave emanating from the atomic nucleus. We will liken this wave to the electric wave sent out by a radio aerial.

Planck’s law, $E = h\nu$, holds good for these matter waves, too; it correlates their energy, E, with their frequency, ν. In this case, E is the energy of the beta particle, or more exactly, that portion of the decay energy which falls to the share of the electron. If, then, the decay probability increases as the energy, the higher is the magnitude ν, the greater will be the decay probability, and therefore the shorter the wavelength of the radiation. If the dipole moment of a vibrating aerial is kept constant which can be achieved by maintaining the peak voltage between the condenser plates of the oscillatory circuit constant, then the radiation will be all the stronger the higher the frequency in the